Producer Theory

Compiled by

Chris Mbukwa

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Production

- A firm uses a technology or production process to transform inputs or factors of production into outputs. Firms use many types of inputs. Most of these inputs can be grouped into three broad categories:
  - **Capital (K).** Long-lived inputs such as land, buildings (factories, stores), and equipment (machines, trucks)
  - **Labor (L).** Human services such as those provided by managers, skilled workers (architects, economists, engineers, plumbers), and less-skilled workers (custodians, construction laborers, assembly-line workers)
  - **Materials (M).** Raw goods (oil, water, wheat) and processed products (aluminum, plastic, paper, steel)
- The output can be a service, such as an automobile tune-up by a mechanic, or a physical product, such as a computer chip or a potato chip

THE SHORT-RUN THEORY OF PRODUCTION

The cost of producing any level of output will depend on the amount of inputs (or ‘factors of production’) used and the price the firm must pay for them. Let us first focus on the quantity of factors used.

Output depends on the amount of resources and how they are used.

Different amounts and combinations of inputs will lead to different amounts of output. If output is to be produced efficiently, then inputs should be combined in the optimum proportions.

Short- and long-run changes in production

- If a firm wants to increase production, it will take time to acquire a greater quantity of certain inputs. For example, a manufacturer can use more electricity by turning on switches, but it might take a while to obtain and install more machines, and longer still to build a bigger, or a second, factory
- If the firm wants to increase output relatively quickly, it will only be able to increase the quantity of certain inputs. It can use more raw materials and more fuel. It may be able to use more labour by offering overtime to its existing work force, or by recruiting extra workers if they are available. But it will have to make do with its existing buildings and most of its machinery.

Short- and long-run changes in production

- The distinction we are making here is between fixed factors and variable factors. A fixed factor is an input that cannot be increased within a given time period (e.g. buildings). A variable factor is one that can.
- The distinction between fixed and variable factors allows us to distinguish between the short run and the long run.
- **The short run is a time period during which at least one factor of production is fixed. Output can be increased only by using more variable factors.**
- For example, if a coffee bar became more successful it could serve more customers per day in its existing shops, if there was space. It could increase the quantity of milk and coffee beans it purchases. It may be able to hire more staff, depending on conditions in the local labour market, and purchase additional coffee machines if there was space to install them. However, in the short run it could not extend its existing shops or have new ones built. This would take more time.

Short- and long-run changes in production

- **The long run is a time period long enough for all inputs to be varied. Given enough time, a firm can build additional factories and install new plant and equipment; a coffee shop can have new shops built.**
- The actual length of the short run will differ from firm to firm and industry to industry. It is not a fixed period of time.
- It might take a farmer a year to obtain new land, buildings and equipment; if so, the short run is any time period up to a year and the long run is any time period longer than a year.
- If it takes a mobile phone handset manufacturer two years to get a new factory built, the short run is any period up to two years and the long run is any period longer than two years.

Definitions

- **Rational producer behaviour** When a firm weighs up the costs and benefits of alternative courses of action and then seeks to maximise its net benefit.
- **Theory of the firm** The analysis of pricing and output decisions of the firm under various market conditions, assuming that the firm wishes to maximise profit.
- **Fixed factor** An input that cannot be increased in supply within a given time period.
- **Variable factor** An input that can be increased in supply within a given time period.
- **Short run** The period of time over which at least one factor is fixed.
- **Long run** The period of time long enough for all factors to be varied

PRODUCTION FUNCTION

- The Production Function
  - The *production function* shows the relationship between quantity of inputs used to make a good and the quantity of output of that good.
- Marginal Product
  - The *marginal product* of any input in the production process is the increase in output that arises from an additional unit of that input.

The short-run production function: total physical product

Let us now see how the law of diminishing returns affects the total output of a firm in the short run. When a variable factor is added to a fixed factor the total output that results is often called total physical product (TPP).

The relationship between inputs and output is shown in a production function. In the simple case of the farm with only two factors – namely, a fixed supply of land (L n) and a variable supply of farm workers (Lb) – the short-run production function would be

TPP = f(Ln, Lb)

The short-run production function: total physical product

This states that the total physical product (i.e. the output of the farm) over a given period of time is a function of (i.e. depends on) the quantity of land and labour employed.

The total physical output illustrated by the short-run production function shows the maximum output that can be produced by adding more of a variable input to a fixed input,: i.e. it shows points that are all technically efficient.

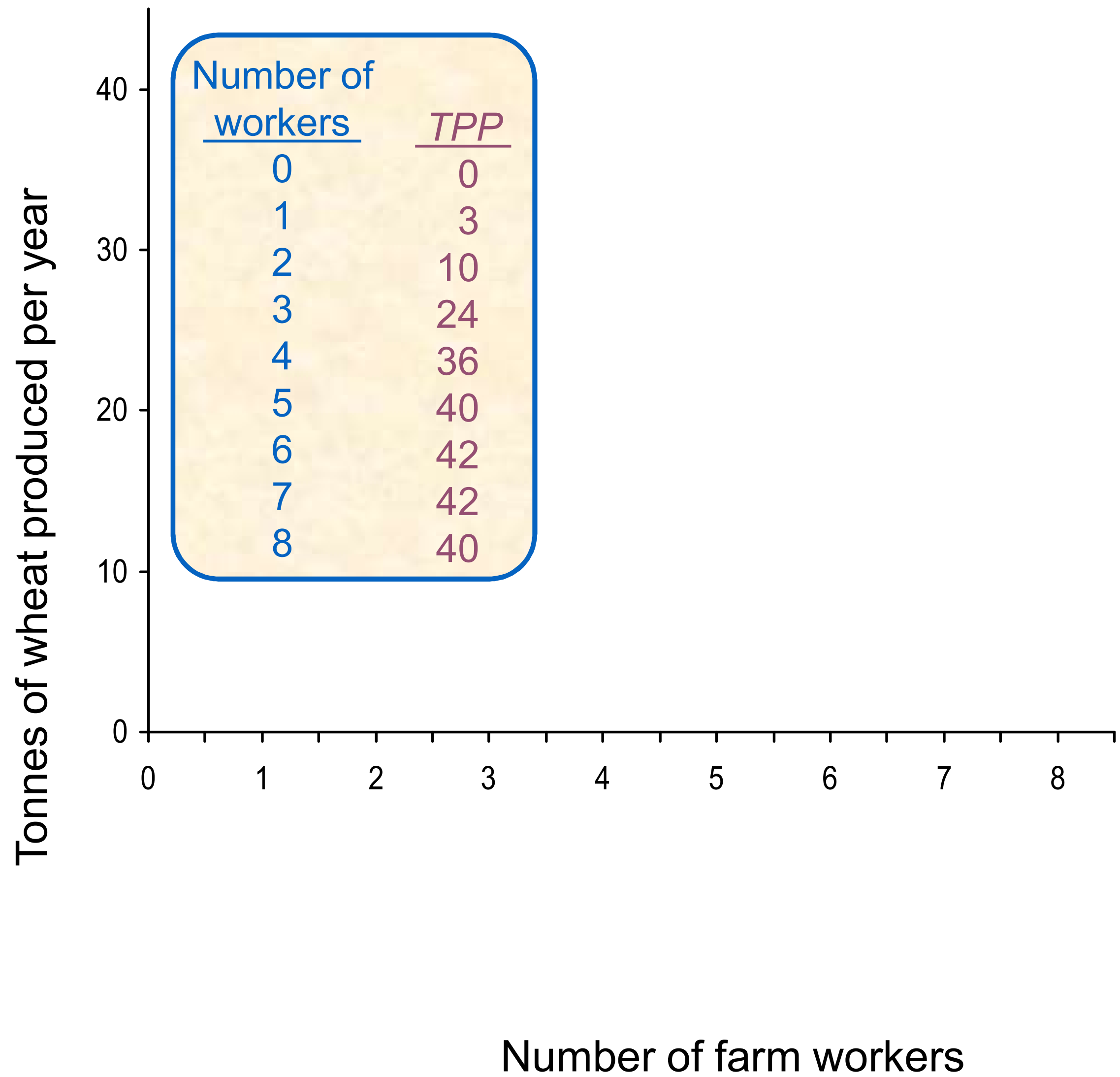
The short-run production function: average and marginal product

- The short-run production function:
  - total physical product (TPP)
  - average physical product (APP)
    - APP = TPP/QV
  - marginal physical product (MPP)
    - MPP = ΔTPP/ΔQV
  - graphical relationship between TPP, APP and MPP

Wheat production per year from a particular farm (tonnes)

Number of workers Lb	TPP	APP	MPP
0	0		
1	3		
2	10		
3	24		
4	36		
5	40		
6	42		
7	42		
8	40		

Wheat production per year from a particular farm



Number of workers	*TPP*
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Wheat production per year from a particular farm

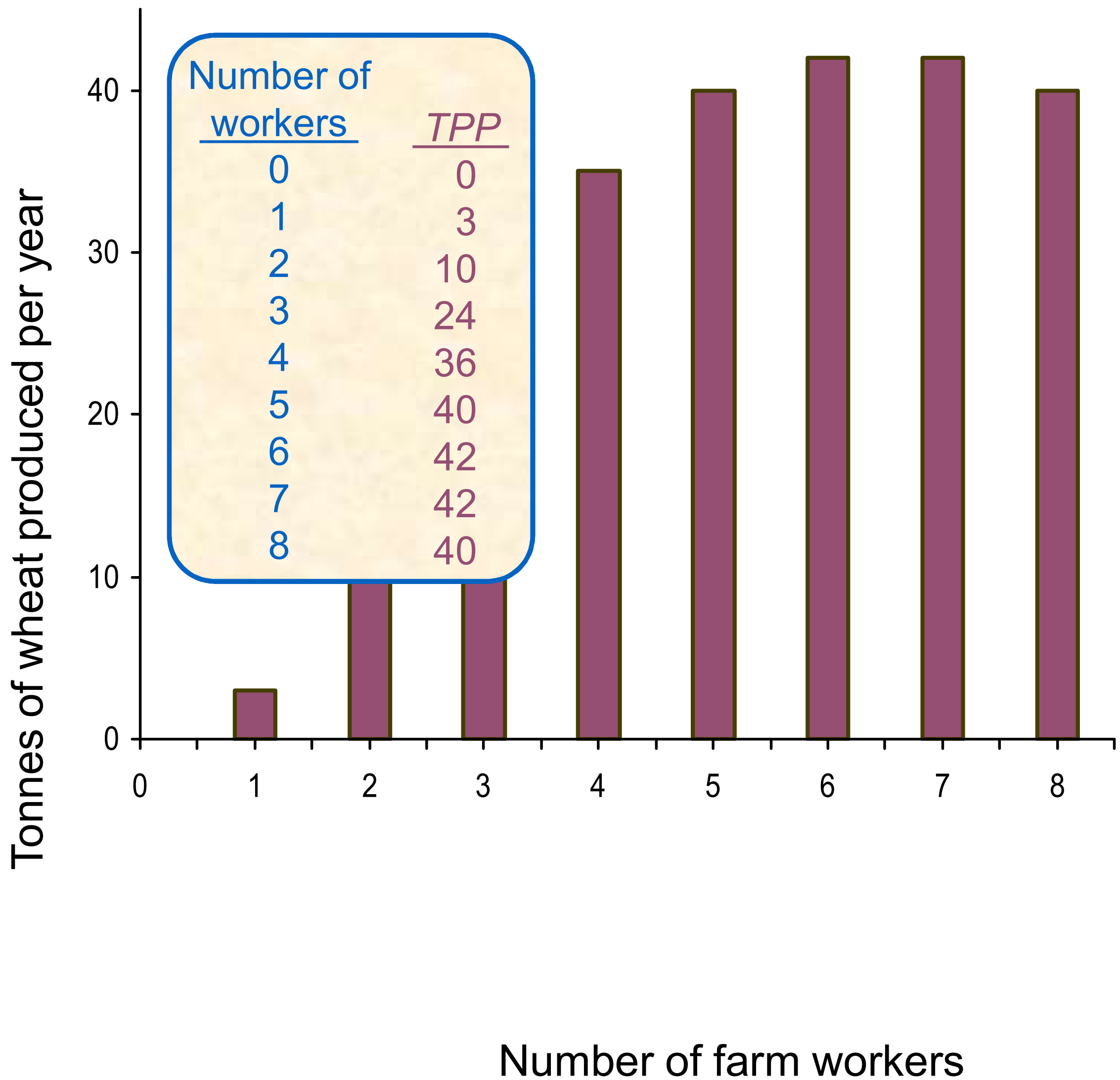
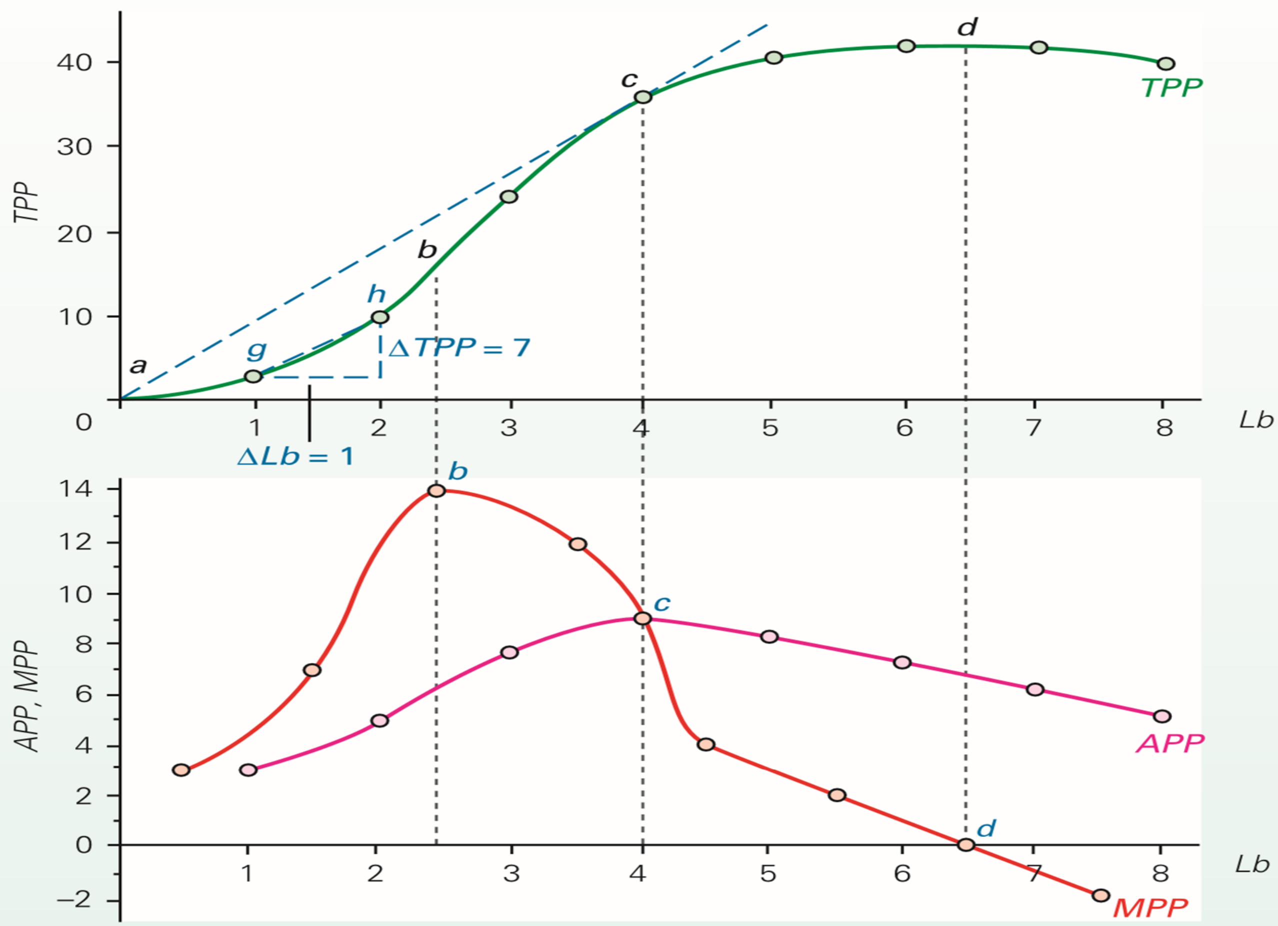


Figure 5.1 Wheat production per year (tonnes)

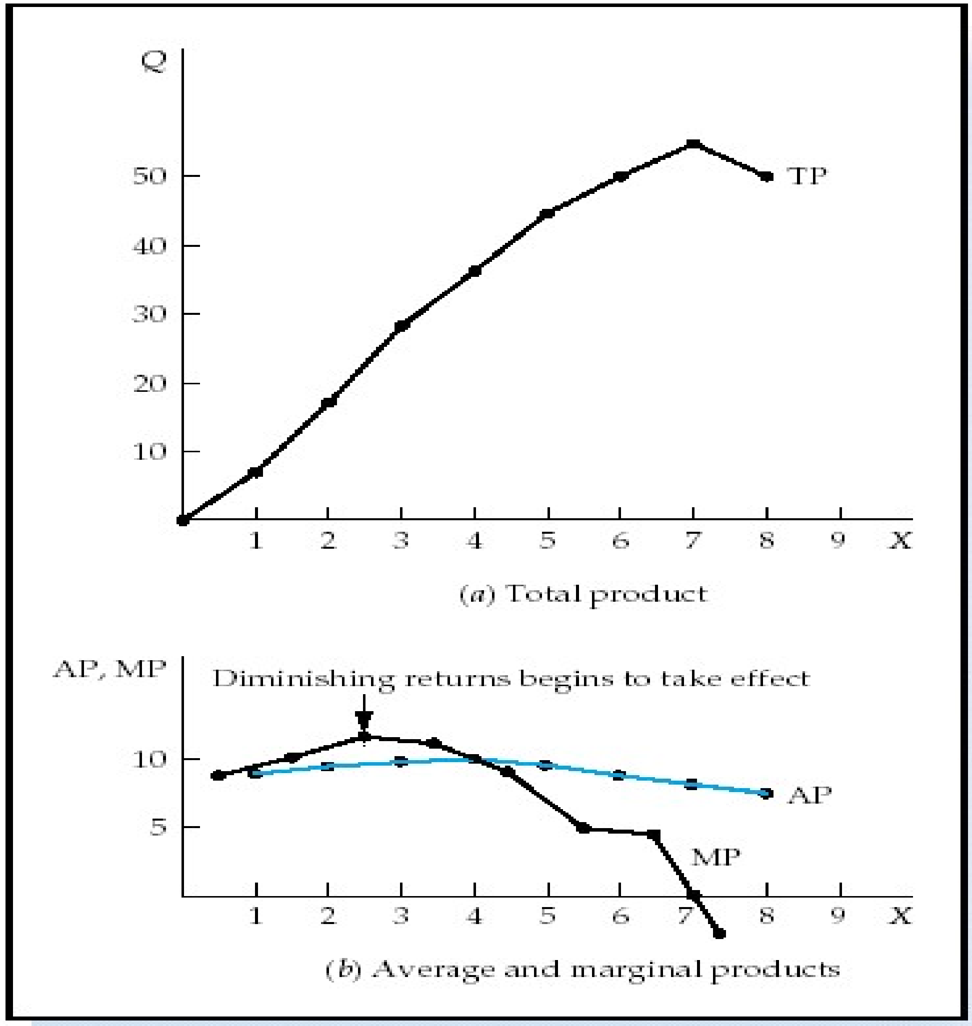


The Production Function

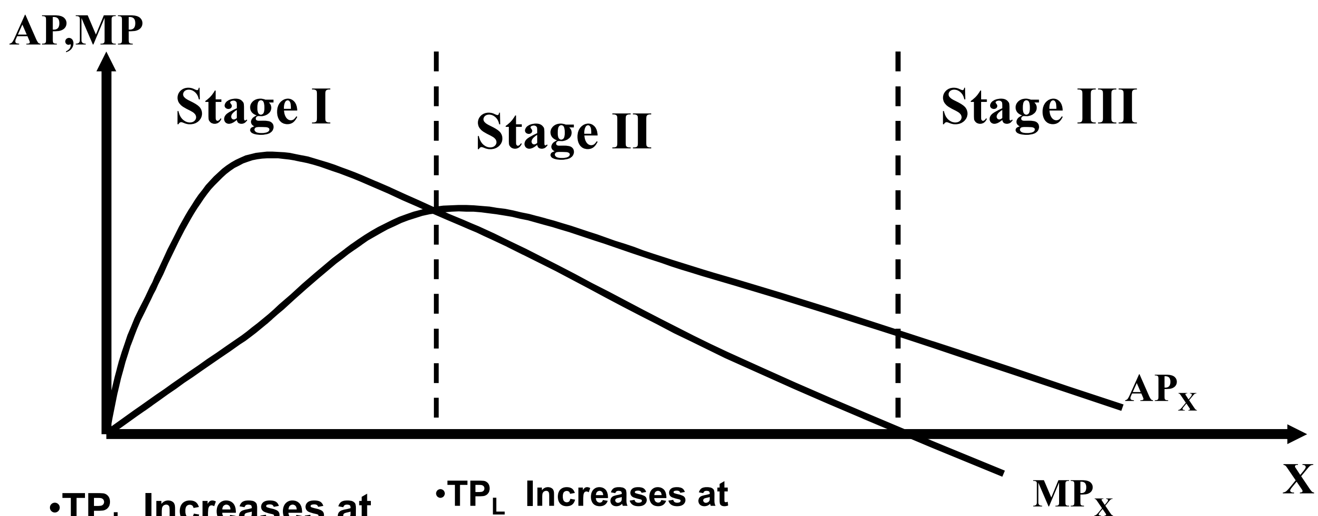
- The **law of diminishing returns** states that when more and more units of a variable input are applied to a given quantity of fixed inputs, the total output may initially increase at an increasing rate and then at a constant rate but it will eventually increases at diminishing rates.
- *Assumptions*. The law of diminishing returns is based on the following assumptions: (i) the state of technology is given (ii) labour is homogenous and (iii) input prices are given.

Short-Run Analysis of Total, Average, and Marginal Product

- If MP > AP then AP is rising
- If MP < AP then AP is falling
- MP = AP when AP is maximized
- TP maximized when MP = 0



Three Stages of Production in Short Run



- TP_L Increases at increasing rate.
- MP Increases at decreasing rate.
- AP is increasing and reaches its maximum at the end of stage I

- TP_L Increases at Diminshing rate.
- MP_L Begins to decline.
- TP reaches maximum level at the end of stage II, MP = 0.
- AP_L declines

- TPL begins to decline
- MP becomes negative
- AP continues to decline

Three Stages of Production				
				Stages
Labor	Total	Average	Marginal	of
Unit	Product	Product	Product	Production
(X)	(Q or TP)	(AP)	(MP)	
1	24	24	24	
2	72	36	48	I
3	138	46	66	Increasing
4	216	54	78	Returns
5	300	60	84	
6	384	64	84	
7	462	66	78	
8	528	66	66	II
9	576	64	48	Diminishing
10	600	60	24	Returns
11	594	54	-6	III
12	552	46	-42	Negative Returns

Definitions

- **Law of diminishing (marginal) returns** When one or more factors are held fixed, there will come a point beyond which the extra output from additional units of the variable factor will diminish.
- **Total physical product** The total output of a product per period of time that is obtained from a given amount of inputs.
- **Production function** The mathematical relationship between the output of a good and the inputs used to produce it. It shows how output will be affected by changes in the quantity of one or more of the inputs used in production holding the level of technology constant.
- **Technical efficiency** The firm is producing as much output as is technologically possible given the quantity of factor inputs it is using

Section summary

1. A production function shows the relationship between the amount of inputs used and the amount of output produced from them (per period of time).
2. In the short run it is assumed that one or more factors (inputs) are fixed in supply. The actual length of the short run will vary from industry to industry.
3. Production in the short run is subject to diminishing returns. As greater quantities of the variable factor(s) are used, so each additional unit of the variable factor will add less to output than previous units: marginal physical product will diminish and total physical product will rise less and less rapidly.
4. As long as marginal physical product is above average physical product, average physical product will rise. Once MPP has fallen below APP , however, APP will fall.

Exercise

The total physical product of a variable factor (e.g. fertilizer) can be expressed as an equation. For example

$TPP = 100 + 32Qf + 10\ Qf^2 – Qf^3$

where TPP is the output of grain in tonnes per hectare, and Qf is the quantity of fertiliser applied in kilograms per hectare.

From this we can derive

1. The APP function
2. The MPP function
3. From the three equations, derive the table if Q ranges from 1 to 9 kgs